INCH POUND

MIL-C-70473 (AR)
AMENDMENT 4
30 MARCH 1992
SUPERSEDING
AMENDMENT 3
17 MARCH 1992

MILITARY SPECIFICATION

CUSHION COMPONENTS FOR METAL CONTAINER FOR CARTRIDGE, 120 mm, TANK AMMUNITION

This Amendment forms a part of Military Specification MIL-C-70473 (AR), dated 8 September 1986, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
2	2
2a	None
3	3
*4	4
5	5
6	6
7	7
8	8
9	9
10	10
11a	None
12	12

* Indicates specific pages revised by this Amendment.

AMSC N/A

FSC-8140

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Custodian:
Army (AR)

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. - The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

	9390367	Sleeve, Cartridge Case
*	9390375	Cushion, Nose, Foam
	9390377	Support, Projectile, Molded
*	12597644	Cushion, Base, Foam
*	12598331	Support, Projectile, Molded
	12631091	Spacer, Cushion, Foam

(Copies of specifications, standards and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951	Commercial Packaging, Standard Practice for

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this

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specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 Materials. - Materials and parts shall be in accordance with applicable drawings and specifications.

* 3.2 Components. - The components shall comply with all requirements specified on drawings 9390375, 9390367, 9390377, 12631091, 12597644 or 12598331 and the drawings and specification referenced thereon.

3.3 Water absorption. - Water absorption of the polyethylene foam parts, at time of packing, shall not exceed 0.08 pounds per square foot of cut surface when tested in accordance with PPP-C-1752.

3.4 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. - The components shall be regular, smooth and free from any creases, wrinkles, cracks, rips, tears, dirt and any other defect which might offset the serviceability, durability, safety and appearance of the assembly.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.1.1 Responsibility for compliance. - All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of ten (10) each of every component.

4.3.2 Inspections to be performed. - See MIL-A-48078.

4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality conformance inspection.

* 4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of cushion components shall contain material from lots with the same interfix number from one manufacturer. Components manufactured with or without laminations for the Cushion, Base, Foam P/N 12597644 or Cushion, Nose, Foam P/N 9390375 shall not be mixed in one lot.

4.4.2 Examination. - See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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TABLE I. - First Article Inspection.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE: Components		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
*	Cushion, Nose, Foam (Drawing 9390375)				
	Examination for defects	10		3.2	4.4.2.2
	Water absorption	10		3.3	4.5.1
*	Cushion, Base, Foam (Drawing 12597644)				
	Examination for defects	10		3.2	4.4.2.2
	Water absorption	10		3.3	4.5.1
	Sleeve, Cartridge Case (Drawing 9390367)				
	Examination for defects	10		3.2	4.4.2.1
	Water absorption	10		3.3	4.5.1
*	Support Projectile, Molded (Drawing 9390377 or 12598331)				
	Examinatrion for defects	10		3.2	4.4.2.3
	Spacer, Cushion Foam (Drawing 12631091)				
	Examination for defects	10		3.2	4.4.2.4

NOTES:

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.1	TITLE Sleeve, Cartridge Case		SHEET 1 OF 1		DRAWING NUMBER 9390367
					NEXT HIGHER ASSEMBLY * 9386832, 9386833 or 12561271
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Material thickness		1.0%	3.2	Gage
* 103	Outside diameter (maximum)		1.0%	3.2	Gage
104	Water absorption	4.4.3.1		3.2	4.5.1
105	Density of material	4.4.3.2		3.2	4.5.2
106	Gaps in thermobond		1.0%	3.2	Visual
* 107	Inside diameter (minimum)		1.0%	3.2	Gage
Minor					
201	Overall length		1.0%	3.2	Gage
202	Poor workmanship		1.0%	3.6	Visual

NOTES:
(a) With each lot of material submitted.

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.2	TITLE * Cushion, Nose, Foam		SHEET 1 OF 1		DRAWING NUMBER 9390375
					NEXT HIGHER ASSEMBLY * 9386832, 9386833 or 12561271
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Thickness		1.0%	3.2	Gage
103	Density of material	4.4.3.2		3.2	Weight Gage
104	Water absorption	4.4.3.1		3.2	4.5.1
Minor					
201	Outside diameter		1.0%	3.2	Gage
202	Number of composite layers		1.0%	3.2	Visual
203	Poor workmanship		1.0%	3.5	Visual

NOTES:
(a) With each lot of material submitted.

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.3	TITLE Support, Projectile, Molded		SHEET 1 OF 2		DRAWING NUMBER * 9390377 or 12598331
					NEXT HIGHER ASSEMBLY * 9386832, 9386833 OR 12561271
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Overall length		1.0%	3.2	Gage
103	Length of penetrator cup		1.0%	3.2	Gage
104	Diameter of sabot cup		1.0%	3.2	Gage
105	Inside diameter of penetrator cup (spike)		1.0%	3.2	Gage
106	Thickness		1.0%	3.2	Gage
107	Outside diameter, end of penetrator cup (spike)		1.0%	3.2	Gage
108	Width of notch		1.0%	3.2	Gage
109	Distance between notches		1.0%	3.2	Gage

NOTES:
(a) With each lot of material submitted.

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.3	TITLE Support, Projectile, Molded		SHEET 2 OF 2		DRAWING NUMBER * 9390377 or 12598331
					NEXT HIGHER ASSEMBLY * 9386832, 9386833 or 12561271
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Angle between fins	(b)	1.0%	3.2	Certification
202	Recess at bottom of sabot cup	(b)	1.0%	3.2	Gage
203	Radius at entrance to sabot cup	(b)	1.0%	3.2	Gage
204	No sprue or flash		1.0%	3.2	Visual
205	Fin thickness		1.0%	3.2	Gage
206	Location of slots	(b)	1.0%	3.2	Gage
207	Location of notches	(b)			
208	Length of slots		1.0%	3.2	Gage
209	Width of slots		1.0%	3.2	Gage
210	Outside diameter of slots		1.0%	3.2	Gage
211	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:
(b) Three pieces per cavity mold check and after a mold re-work affecting applicable dimensions.

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH 4.4.2.4	TITLE Spacer, Cushion, Foam		SHEET 1 OF 1		DRAWING NUMBER 12631091
					NEXT HIGHER ASSEMBLY * 12630717, 12630718 or 12561271
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Density	(a)		3.2	Certification
Minor					
201	Thickness		1.0%	3.2	Gage
202	Outside diameter		1.0%	3.2	Gage
203	Poor workmanship		1.0%	3.5	Visual

NOTES:

(a) With each lot of material submitted.

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH * 4.4.2.6	TITLE Cushion, Base, Foam		SHEET 1 OF 1		DRAWING NUMBER 12597644
					NEXT HIGHER ASSEMBLY 9386832 or 9386833
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material certification	(a)		3.2	Certification
102	Thickness		1.0%	3.2	Gage
103	Density of material	4.4.3.2		3.2	Weight Gage
104	Water absorption	4.4.3.1		3.2	4.5.1
Minor					
201	Outside diameter		1.0%	3.2	Gage
202	Number of composite layers		1.0%	3.2	Visual
203	Poor workmanship		1.0%	3.5	Visual

NOTES:
(a) With each lot of material submitted.

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4.4.3 Testing.

* 4.4.3.1 Water absorption. - (See drawings 9390375, 9390367 and 12597644) Major Defect. Three samples per material lot shall be selected prior to start of production. If any sample fails to meet the requirements, the lot shall be rejected.

* 4.4.3.2 Density. - (See drawings 9390375, 9390367 and 12597644) Major Defect. Three samples per material lot shall be selected prior to start of production. If any sample fails to meet the requirements, the lot shall be rejected.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.5 herein.

4.5 Methods of inspection.

4.5.1 Water absorption. - The water absorption shall be determined in accordance with PPP-C-1752.

4.5.2 Density. - The density shall be determined in accordance with PPP-C-1752.

5.0 PACKAGING

5.1 Preservation and packaging. - None required.

5.2 Packing.

5.2.1 Level A. - The cushion components shall be packed in accordance with Level A of MIL-STD-1169.

5.2.2 Level B. - The cushion components shall be packed in accordance with Level B of MIL-STD-1169.

5.2.3 Commercial. - The cushion components shall be packed in accordance with ASTM D3951.

5.3 Marking.

5.3.1 Level A. - The cushion components shall be marked in accordance with Level A of MIL-STD-1169.

5.3.2 Level B. - The cushion components shall be marked in accordance with Level B of MIL-STD-1169.

5.3.3 Commercial. - The cushion components shall be marked in accordance with ASTM D3951 and the following additional instructions.

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